MAGNETIC EFFECT OF ELECTRIC CURRENT

PYQ • 2021 - 2026

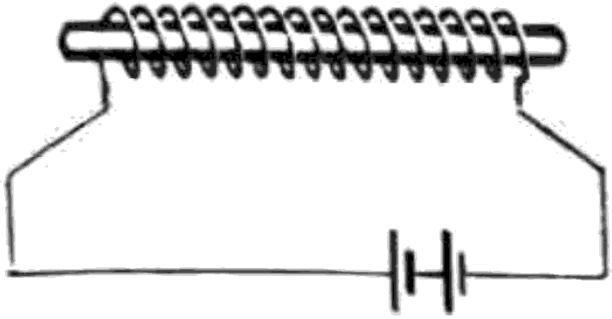
1. 2026 **3 MARKS**

Observe the figure of a current carrying solenoid.

(a) What is meant by a solenoid ?

(b) The end connected to the positive terminal of the cell will always be the North pole of a solenoid. Is this statement correct ? Justify your answer.

(c) Write any two ways to increase the strength of the magnetic field formed around a current carrying solenoid.



2. 2026 **2 MARKS**

Observe the figure showing a rule related to electricity and magnetism. Which law does this figure represent ? State the rule.



3. 2026 **1 MARK**

Match the following and choose the correct option.

A. Bar magnet B. Magnetism is permanent C. Polarity can be changed

D. Current carrying solenoid E. Magnetic strength can be varied F. Polarity cannot be changed

(a) AEC and DBF (b) ABC and DEF

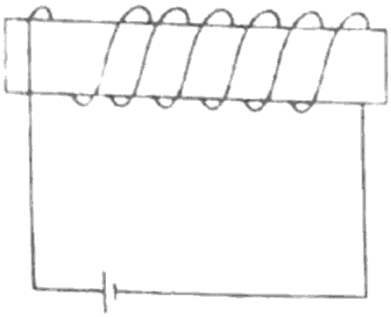
(c) ABF and DEC (d) AEF and DBC

4. 2025 **3 MARKS**

Observe the diagram of a current carrying solenoid.

(a) Copy the figure and mark the direction of current flow through the coil.

(b) Identify and mark the north pole of this solenoid. Justify your answer.



5. 2025 **1 MARK**

Which of the following statements correctly describes the working of a loudspeaker ?

(a) Sound energy is converted into electrical energy based on the motor principle.

(b) Electrical energy is converted into sound energy based on the motor principle.

(c) Electrical energy is converted into sound energy based on electromagnetic induction.

(d) Sound energy is converted into electrical energy based on electromagnetic induction.

6. 2025 **1 MARK**

In Right Hand Thumb Rule, the thumb indicates the direction of _____ .

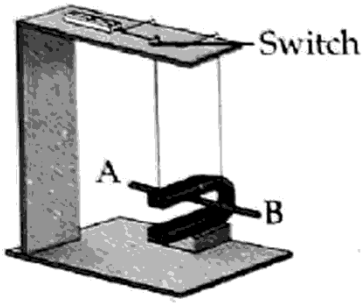
7. 2024 **2 MARKS**

A conductor AB suspended freely between the poles of a U magnet is depicted. DC source is connected to the circuit.

(a) State the rule to find the direction of motion of the conductor AB when a current is switched on.

(b) Write two methods to reverse the direction of motion of the conductor.

source is connected to the circuit



8. 2024 **2 MARKS**

A magnetic field is formed around a current carrying conductor.

(a) Write the law to find the direction of this magnetic field.

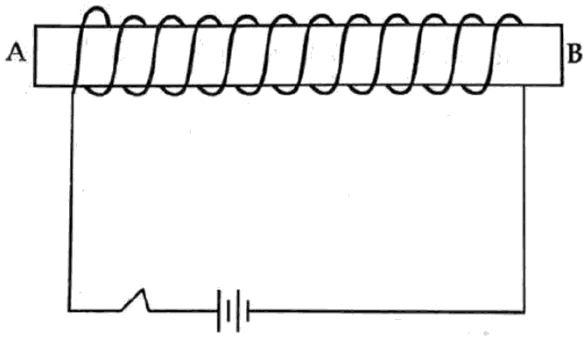
(b) Write two methods to increase the strength of magnetic field by current carrying conductors.

9. 2023 **3 MARKS**

A DC current passes through an insulated copper coil wound over a soft iron core as shown in the figure.

(a) By just observing the experimental setup how will you identify the south pole of the electromagnet ?

(b) Compare the properties of an electromagnet and a permanent bar magnet.



10. 2023 **2 MARKS**

A conductor kept in a magnetic field experiences a force when current passes through it.

(a) Name the rule that helps us to find the direction of the force.

(b) Name any two devices that work based on this principle.

11. 2023 **1 MARK**

The prominent effect produced when a current passes through a solenoid is _____ .

[chemical effect, mechanical effect, magnetic effect, light effect]

12. 2022 **3 MARKS**

(a) What is the energy transformation taking place in a moving coil loud speaker ?

(b) Explain the working of a moving coil loud speaker.

13. 2022 **1 MARK**

Which rule helps us to find the direction of motion of a current carrying conductor placed in a magnetic field ?

(Joule's law, Maxwell's right hand thumb rule, Fleming's left hand rule, Fleming's right hand rule)

14. 2021 **2 MARKS**

DC current is flowing through a solenoid AB. The direction of current at the end A is anticlockwise.

(a) Write the polarity at the end A.

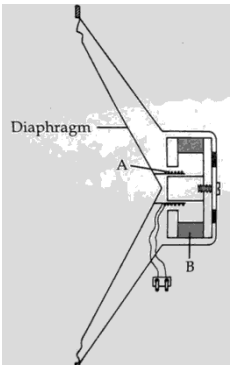
(b) The solenoid is replaced by a current carrying conductor. Give the name of the law that helps to determine the direction of magnetic field produced.

15. 2021 **2 MARKS**

Observe the figure of a moving coil loudspeaker.

(a) Name the parts labeled as A and B.

(b) Write the function of diaphragm in this device.



16. 2021 **2 MARKS**

Write any two methods to increase the magnetic strength of a current carrying solenoid.

17. 2021 **1 MARK**

The non-rotating part in a dc motor is :

[Armature, Split ring, Graphite Brush]